



Time Series Components Transcript

Hello everyone, welcome to lesson 1 video 3. In this lesson, we will learn about components of time series. Recognising these components helps us to separate systematic patterns from random noise. As I told you, we will be decomposing the entire time series into its components.

This entire lecture or this entire video is about how do we decompose a time series into its associated components. So, these are the four key components that a time series is made up of. The very first thing is trend.

This is a long term direction or the movement of the time series. It could increase with time, it could decrease with time, it could have a different pattern altogether. But this particular thing when it increases with time or decreases with time is called as a trend.

You can think of with time as something goes linearly increasing or decreasing that is trend. Then you have seasonality. This is a fixed periodic pattern.

If with time something repeats itself, that's a periodic pattern. That is also one of the key components because you know what is the cyclicity or seasonality of the entire time series. Then cyclicity is also very similar to what seasonality is.

It's a long time repeating effect. Then we have irregularity which means that all those unpredictable noises. When you analyse stock market's data or shares, you also see the prices of the shares.

They sometimes go unevenly up, sometimes go unevenly down. Those are noises. Those are something which you actually cannot predict.

These are the things that constitute our entire time series. Let's say for example, if I move to shares of Tesla, you see that there is a proper time series. There is a proper cyclicity in the entire event.

Now in this, we also have certain key moments wherein we cannot predict those unevenness. Let's say for example, there are uneven this entire period right here. These certain things are uneven.

There are certain moments because of which this entire time series data might have irregularities or unevenness. We have trend, we have cyclicity or seasonality but we can also have these sharp edges which is very unpredictable in nature. That is why we call them as irregularity.

Now let's say for example, this is our original time series data. Now you can easily see that there is some sort of cyclic pattern and that's what is decomposed here. Now there is also an increasing trend as well because from here it is going upwards which is what is captured by trend component.



Then you have this irregular component which measures the uneven sharpness. If you see at some places, you have uneven sharpness generated. These are those uneven irregularities in the entire time series.

Once we have broken down this entire time series data, or this original time series into trend, seasonal and irregularities, we can actually use all of these components to demystify something in what is going to happen in the future because we can predict using the trend what is the next trend. We can predict using the seasonal component what is the next value it is likely to be observed and we can also use average of irregularities to find out what is the average irregularity that we will be getting and then we can combine all of these three to predict the future value of the time series. So that's the simple notion of the entire predictability in the time series.

Now combining the understandings that we have two different types of models that we built. One is called as additive and the other one is called as multiplicative model. Additive model is when you add the components we have studied in the previous slide, trend, seasonal and you have irregularities.

When you add them together, this becomes your additive model. When you multiply them together, this becomes your multiplicative model. So basically you have two different types of models.

One is called as your additive model. The second is called as your multiplicative models. In this type of model, what might happen is the swings that you get might be bigger in terms of your absolute numbers.

The trend goes up and up. In this type of a model, there is more linear trend. The swings don't get bigger as the trend goes up.

So if you think of it, this is almost an additive time series because you do not have, you have the same differencing between the cycles. Let's say for example, the cycle is equally repeating in terms of the time. If the cycle is repeating but the cycle gap also increases, that is multiplicative time series.

That's what is indicated here. I hope this was all clear to everyone. In summary, every time series can be thought of a combination of trend, seasonality, cyclic and irregular components.

By underlying or understanding these elements, we can build better forecasting models. In the next lesson, we will explore forecasting using smoothing methods.

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